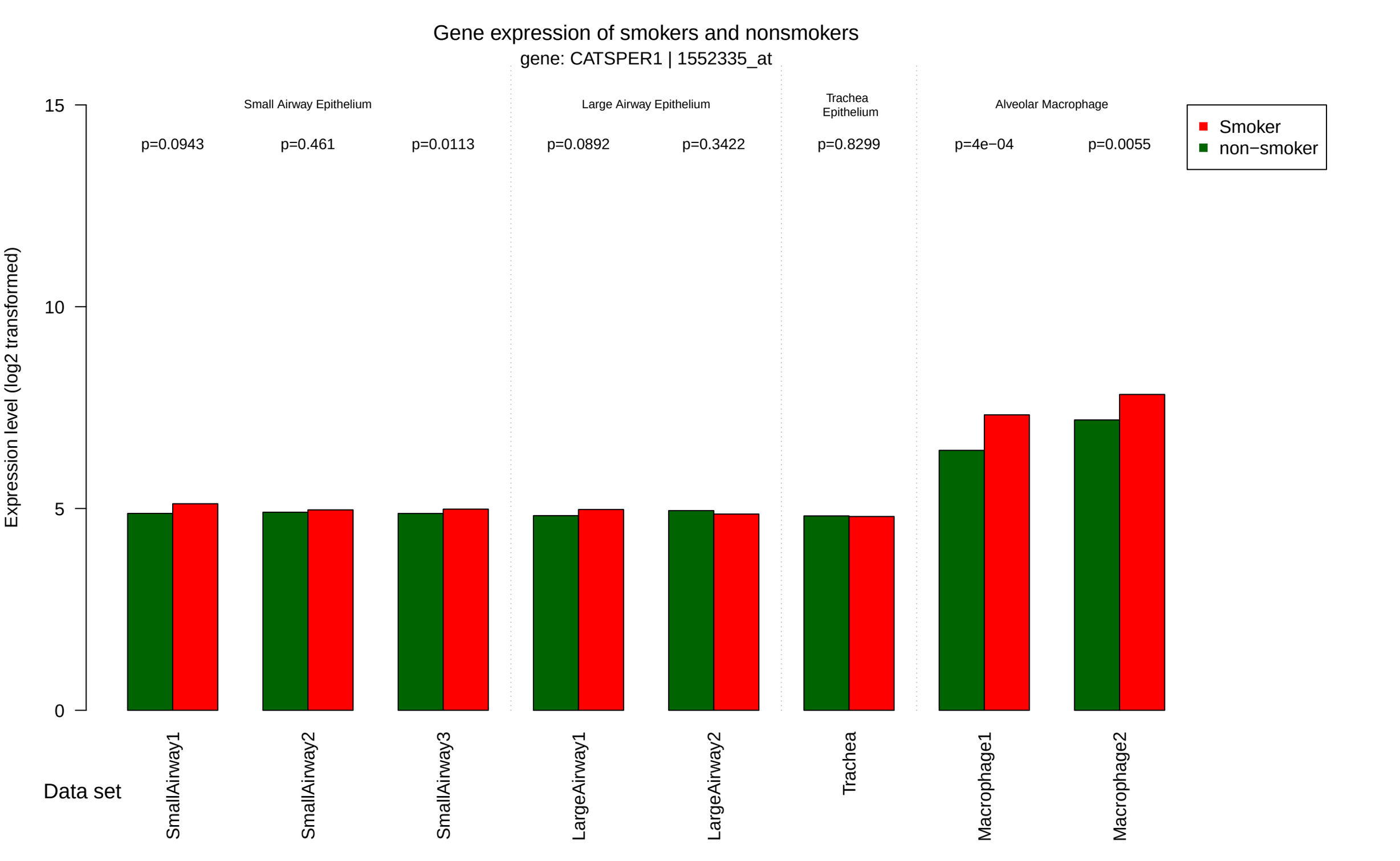
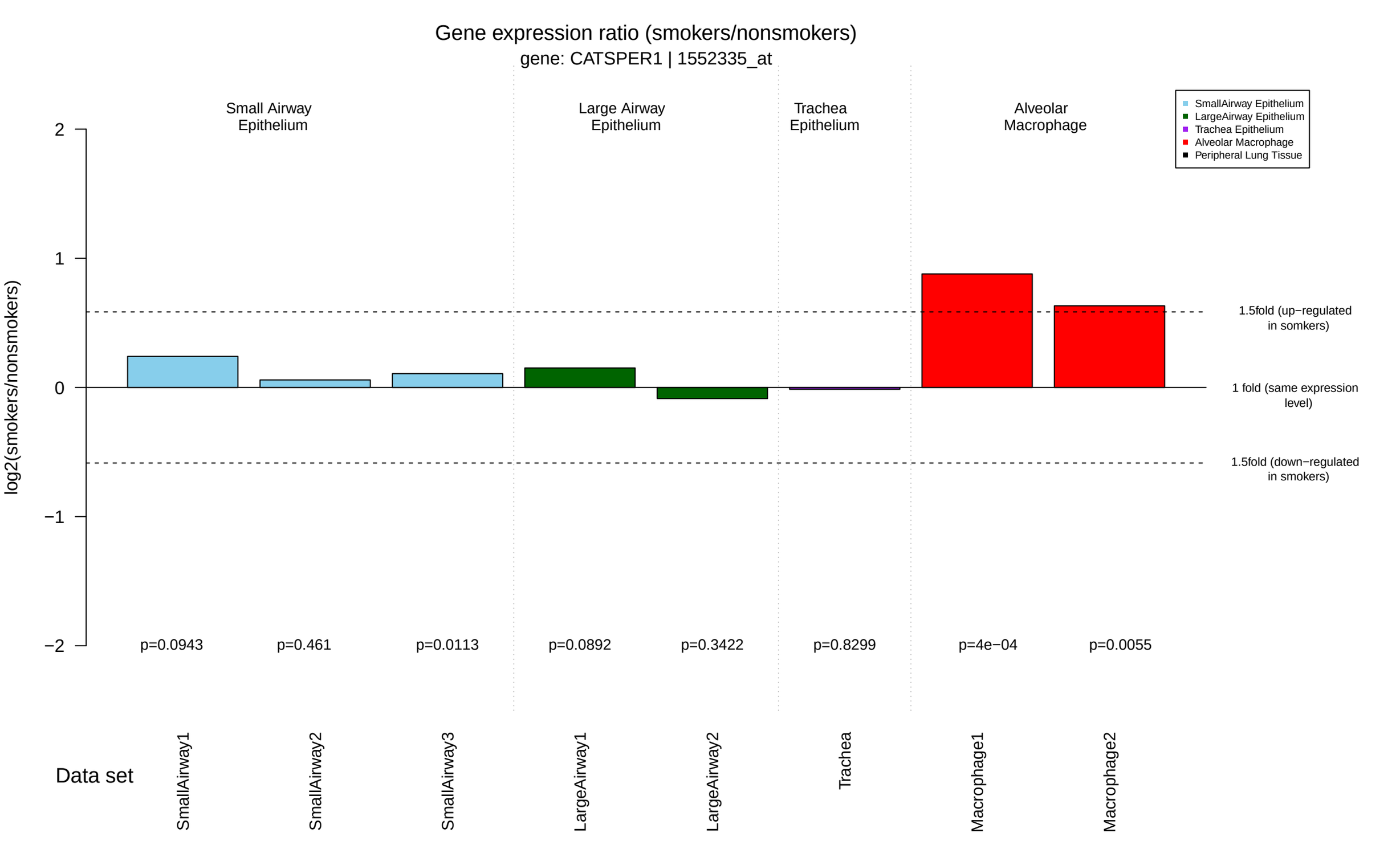
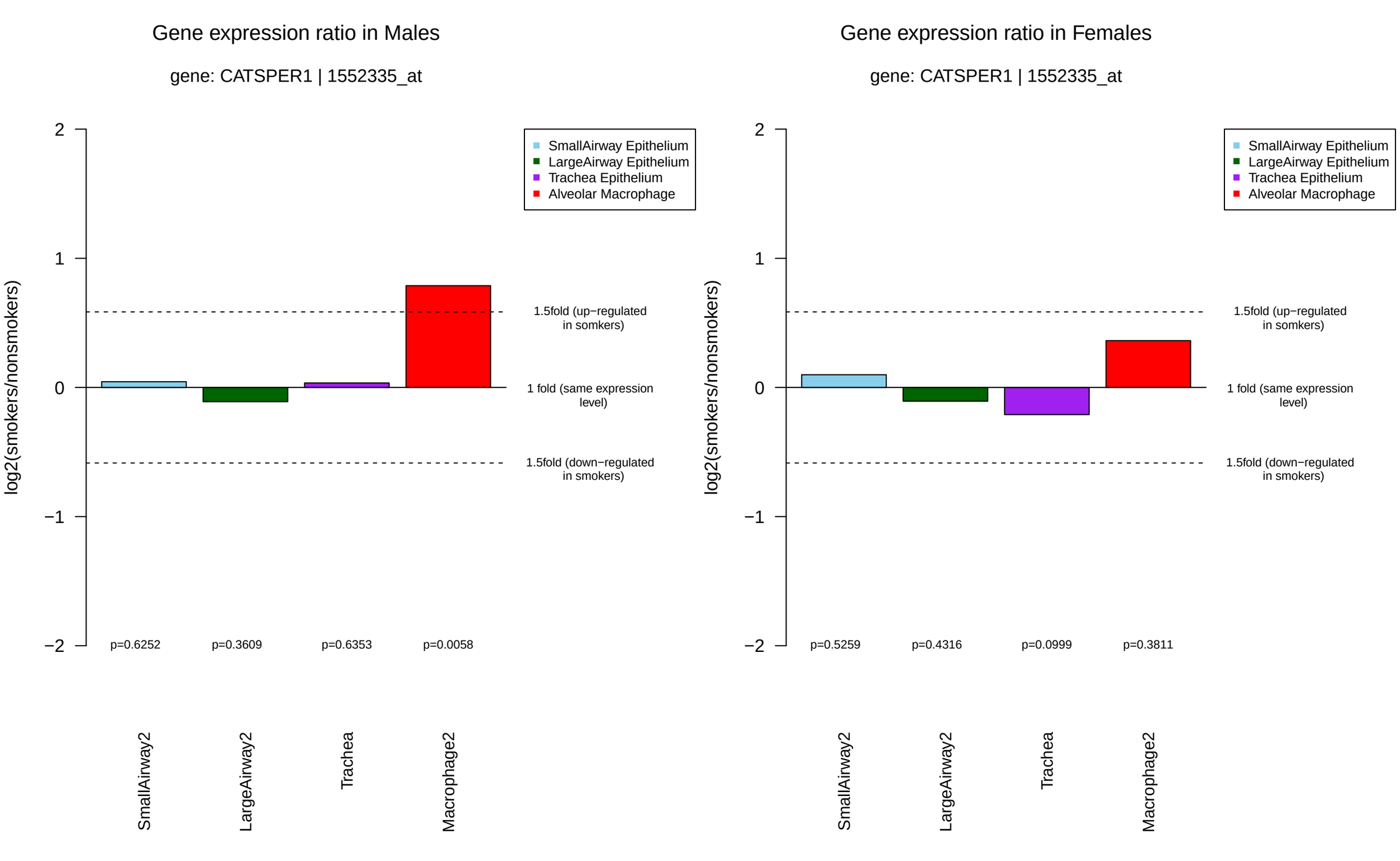
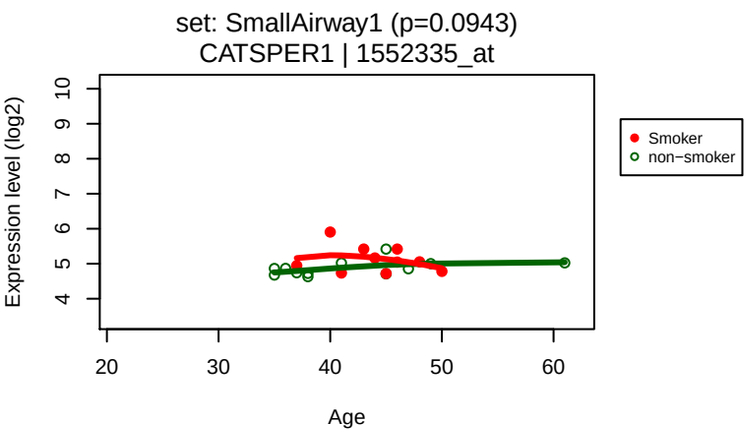
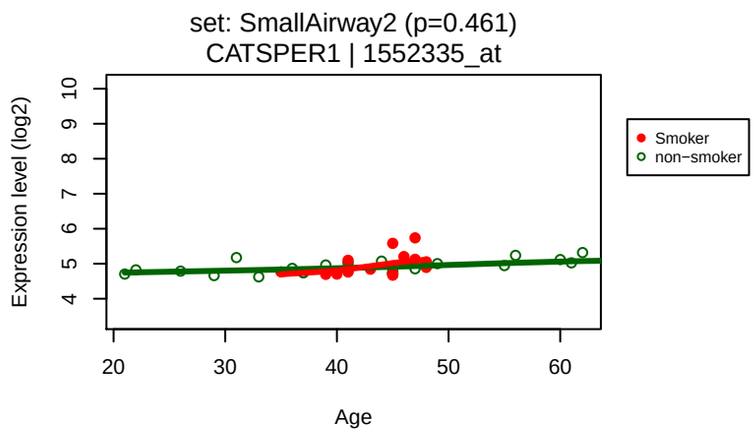
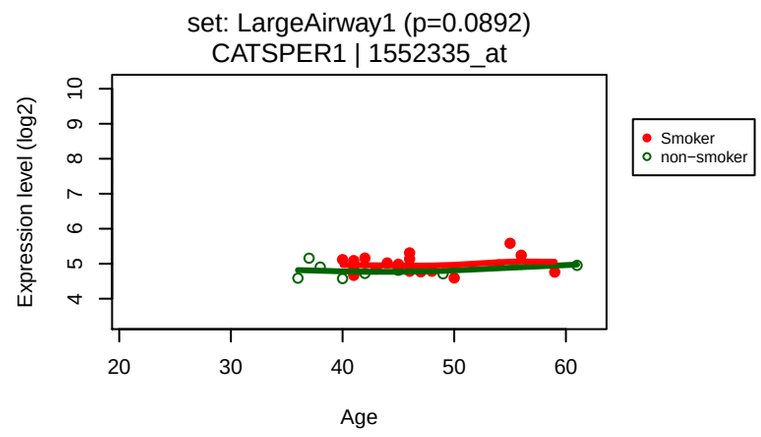

Gene expression of smokers and nonsmokers
gene: CATSPER1 | 1552335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0943
p=0.461
p=0.0113
p=0.0892
p=0.3422
p=0.8299
p=4e−04
p=0.0055
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

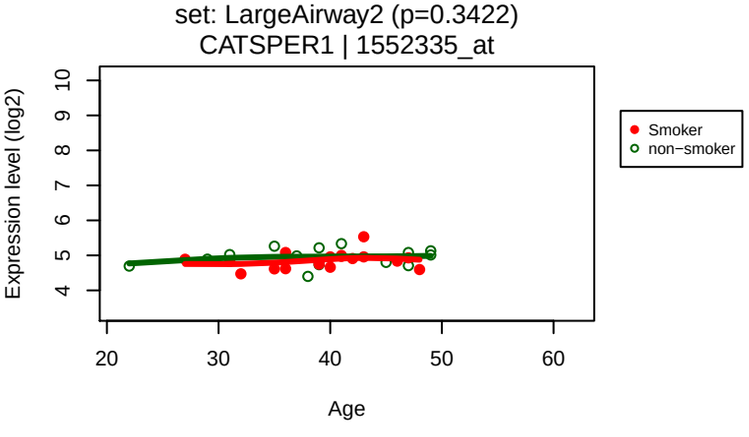

Gene expression ratio (smokers/nonsmokers)
gene: CATSPER1 | 1552335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0943
p=0.461
p=0.0113
p=0.0892
p=0.3422
p=0.8299
p=4e−04
p=0.0055
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

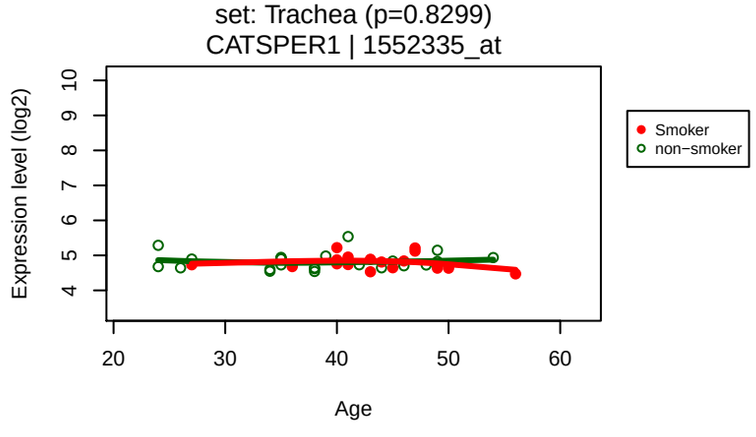

Gene expression ratio in Males
gene: CATSPER1 | 1552335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6252
p=0.3609
p=0.6353
p=0.0058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CATSPER1 | 1552335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5259
p=0.4316
p=0.0999
p=0.3811
SmallAirway2
LargeAirway2
Trachea
Macrophage2

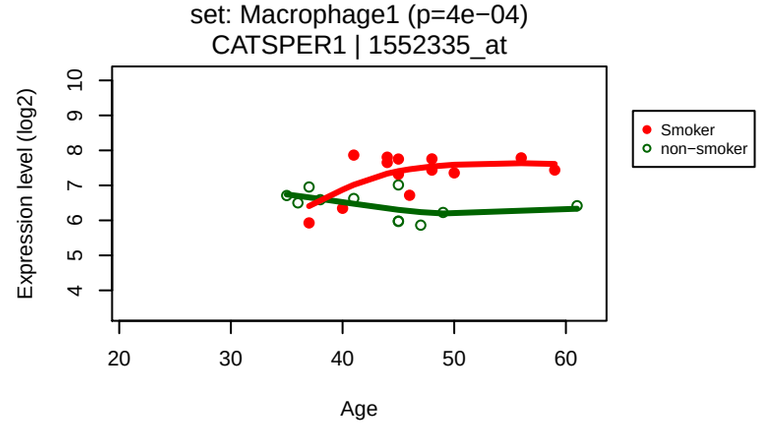

set: SmallAirway1 (p=0.0943)
CATSPER1 | 1552335_at
Expression level (log2)
Age
Smoker
non−smoker

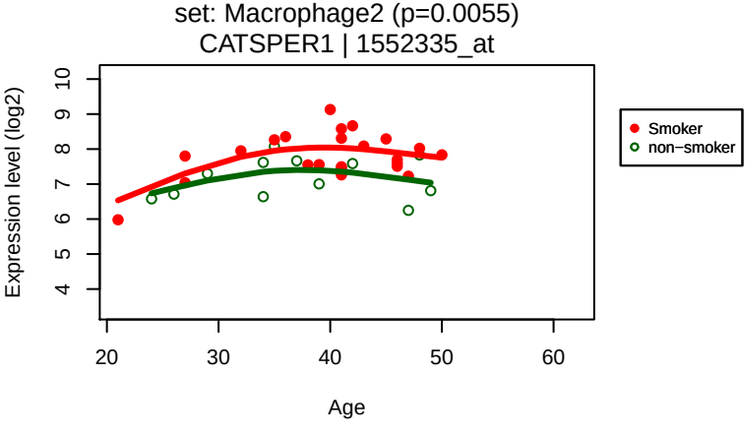

set: SmallAirway2 (p=0.461)
CATSPER1 | 1552335_at
Expression level (log2)
Age
Smoker
non−smoker

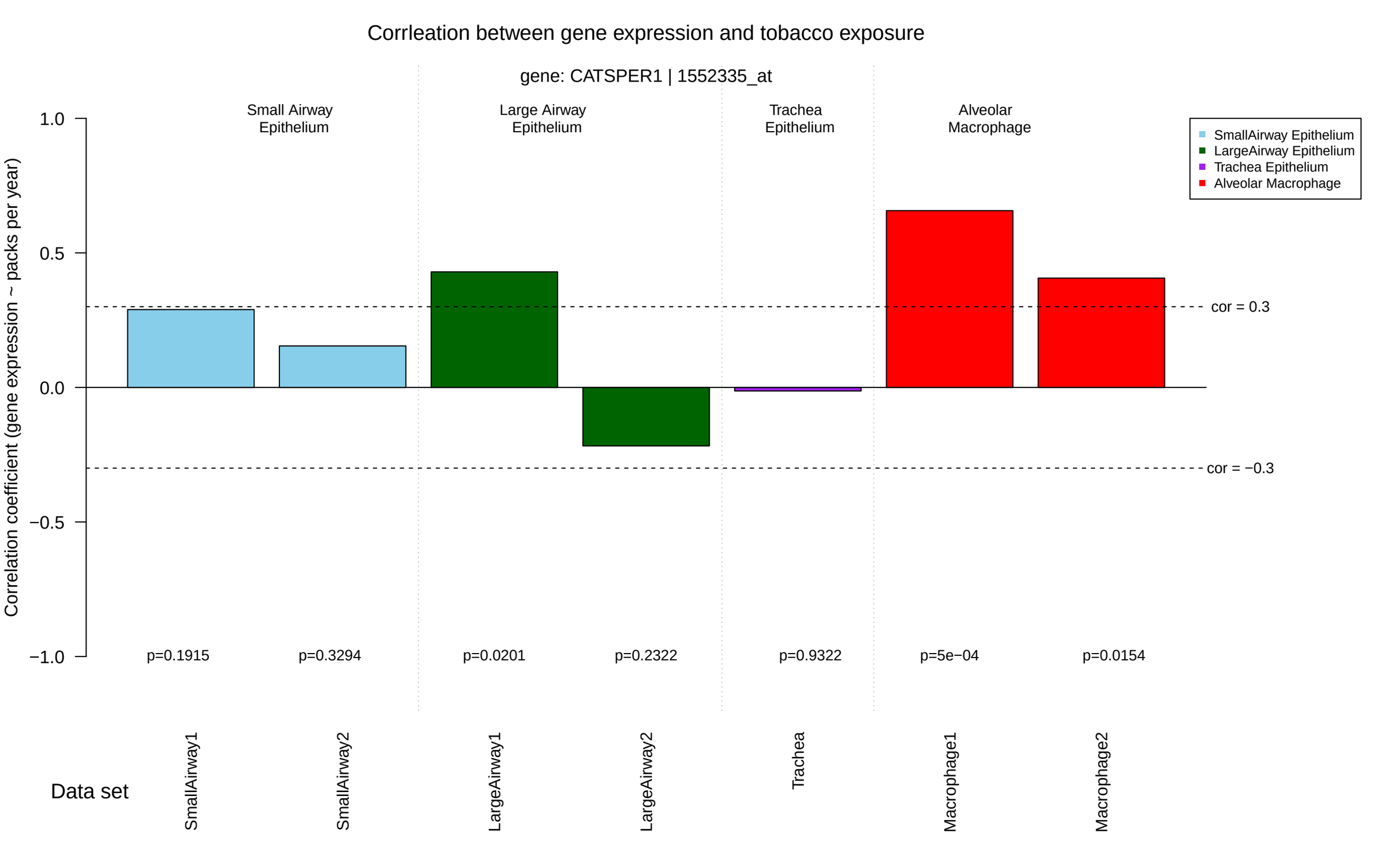

set: LargeAirway1 (p=0.0892)
CATSPER1 | 1552335_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3422)
CATSPER1 | 1552335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8299)
CATSPER1 | 1552335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=4e−04)
CATSPER1 | 1552335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0055)
CATSPER1 | 1552335_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CATSPER1 | 1552335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1915
p=0.3294
p=0.0201
p=0.2322
p=0.9322
p=5e−04
p=0.0154
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2